

DoorDash S&O Case Study

Delivery Operations — Analyst Summary Report

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Dataset:	~20,000 orders cleaned to 19,346 valid records
Time Period:	One calendar month (UTC timestamps)
Markets:	56 delivery regions
Date:	June 2025

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1. Executive Summary

DoorDash operates a three-sided marketplace connecting Customers (demand), Dashers (supply), and Merchants (inventory). This report analyzes one month of delivery data to surface operational inefficiencies and revenue opportunities. The analysis identifies the driver pickup wait as the dominant bottleneck — averaging 12.9 minutes per order, it consumes 43.5% of total delivery time while restaurant preparation averages just 0.6 minutes. Three prioritized recommendations follow, each supported by data and accompanied by a concrete execution plan.

Metric	Value	Benchmark / Signal
Total Orders	19,346	After cleaning from 20,000 raw
Gross Order Value	\$701,773	~\$36.3K/day average
Avg Order Value	\$36.27	Median \$30.27 — right skewed
Avg Delivery Time	29.6 min	P90 = 46.7 min; P95 = 54.6 min
On-Time Rate	88.3%	11.7% exceed 45-min SLA
Refund Rate	1.4%	3 markets at 3-4× average
Discount % of GOV	1.7%	No AOV lift from discounts
Tip Rate	87.4%	Strong proxy for satisfaction
Repeat Customer Rate	23.9%	76.1% one-time buyers

2. Data Quality & Methodology

Data Cleaning Pipeline

The raw dataset contained 20,000 rows. After systematic cleaning, 19,346 rows (96.7%) were retained for analysis. The following steps were applied:

Step	Issue	Action	Rows
1	ORDER_TOTAL = 0	Excluded (cancelled orders — A2)	430
2	Negative driver_pickup_min	Removed (timestamp error — A4)	3
3	Missing delivery timestamps	Removed (mid-flow cancellations)	221
4	DISCOUNT_AMOUNT < 0	Zeroed out (data entry error — A5)	47
5	total_delivery_min > 240	Capped (extreme outlier filter — A3)	0
	Final clean dataset		19,346

Missing Values Summary

Column	Missing Count	Missing %	Likely Cause
DRIVER_ID	457	2.28%	Driver no-shows / unassigned
DRIVER_AT_RESTAURANT_DATETIME	438	2.19%	Driver no-shows
driver_pickup_min (derived)	438	2.19%	Cascades from above
DELIVERED_TO_CUSTOMER_DATETIME	436	2.18%	Mid-flow cancellations
ORDER_TOTAL	187	0.94%	Cancelled pre-confirmation
PLACED_ORDER_WITH_RESTAURANT_DATETIME	7	0.03%	System/logging error

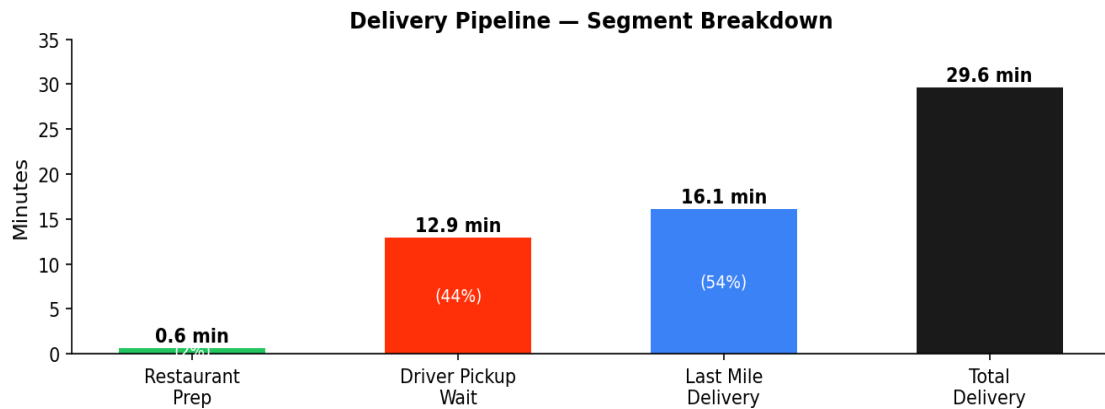
Insight: The 2.2% missing delivery completion rate likely represents driver no-shows or last-minute order cancellations. While low, tracking this cohort over time could reveal supply reliability issues in specific regions or time windows.

Descriptive Statistics — Full Table

Metric	Mean	Median	Std Dev	P25	P75	P90	Skew
ORDER_TOTAL (\$)	36.27	30.27	21.52	22.73	43.26	61.36	2.77
DISCOUNT_AMOUNT (\$)	0.60	0.00	1.69	0.00	0.00	2.22	4.88
TIP_AMOUNT (\$)	3.81	3.50	3.17	2.00	5.00	7.50	2.22
REFUNDED_AMOUNT (\$)	0.18	0.00	2.09	0.00	0.00	0.00	19.20
total_delivery_min	29.57	26.65	15.61	19.48	36.05	46.70	3.21
restaurant_prep_min	0.60	0.00	7.17	0.00	0.02	—	16.94
driver_pickup_min	12.86	10.55	10.17	6.30	17.02	—	3.23
last_mile_min	16.11	14.57	7.98	10.40	20.01	—	1.79

3. Business KPI Dashboard & Delivery Pipeline

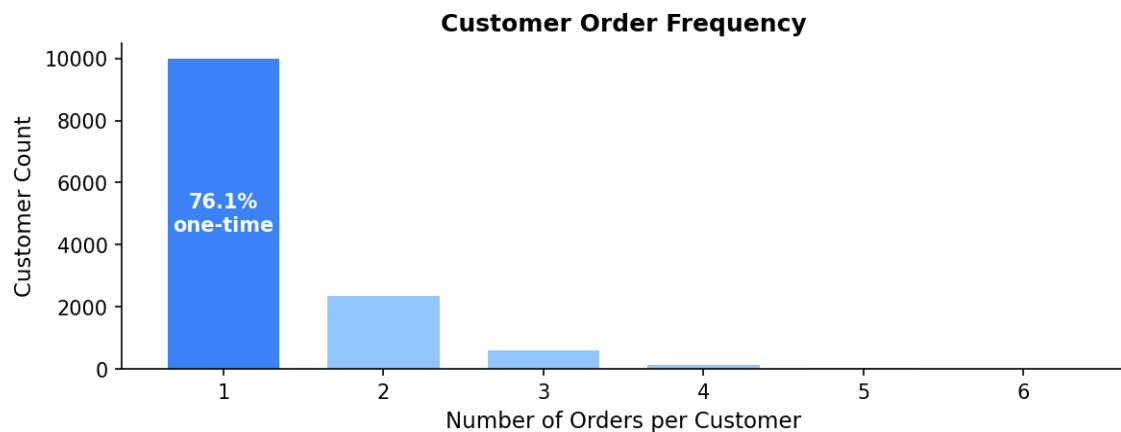
The delivery pipeline has three segments: restaurant preparation, driver pickup wait, and last-mile delivery. The chart below quantifies each contribution to the 29.6-minute average.



Critical Finding: Driver pickup wait (12.9 min, 43.5% of total) dwarfs restaurant prep (0.6 min, 2%). The bottleneck is Dasher dispatch timing, not merchant readiness. A 3-minute reduction in pickup wait would yield a 10% improvement in average delivery time.

4. Customer Analysis

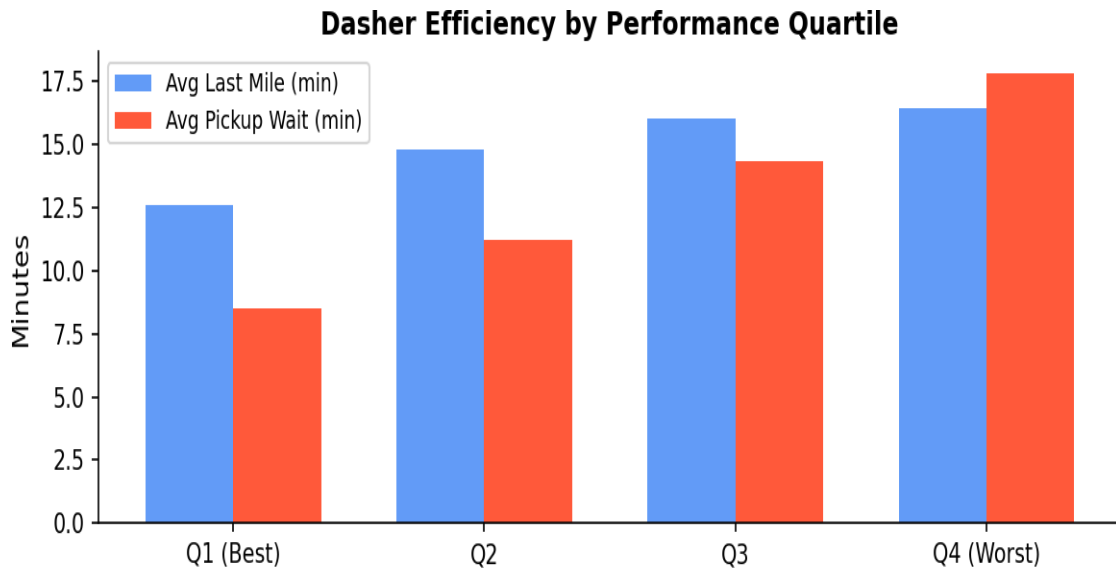
13,092 unique customers placed orders in the sample month. Customer behavior reveals a significant retention gap: 76.1% of customers ordered only once. This single metric represents the largest untapped revenue opportunity in the dataset.



- **One-time buyers:** 76.1% (9,967 customers) — highest priority for retention targeting
- **Repeat buyers:** 23.9% (3,125 customers) — average 2.8 orders each; higher LTV
- **Top 10% by spend:** Each spent \$99.44+ — these are likely DashPass subscribers
- **Refund pattern:** 1.4% of orders resulted in refunds — total refund cost: \$3,507
- **Discount usage:** Discounted orders had lower AOV (\$35.81 vs \$36.42) — negative ROI signal

5. Dasher (Supply) Analysis

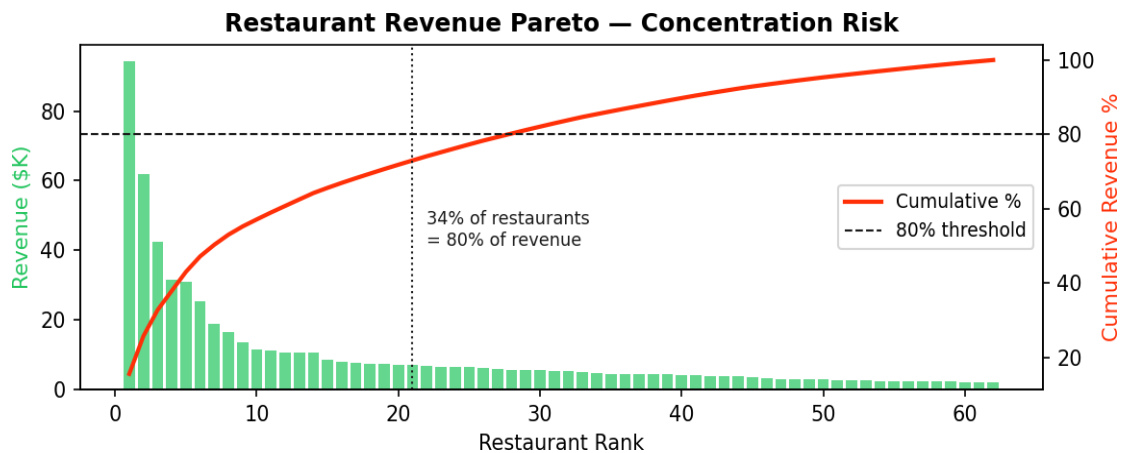
205 Dashers completed 10 or more deliveries in the sample period (filtered for reliability). Performance varies significantly across the quartile distribution, suggesting structured coaching or incentive programs could close the gap.



- **Active Dashers:** 205 with ≥ 10 deliveries; excluded < 10 delivery Dashers (assumption A7)
- **Pickup wait gap:** Best-quartile Dashers average 8.5 min pickup vs 17.8 min for worst (2.1 $\times$ delta)
- **Last-mile gap:** Q1 Dashers: 12.6 min vs Q4: 16.4 min — 3.8-minute efficiency gap
- **Driver no-shows:** 457 orders (2.3%) missing DRIVER_ID — supply reliability concern
- **Implication:** Closing the Q4-to-Q1 gap would reduce avg delivery time by ~2 minutes fleet-wide

6. Merchant Analysis

62 unique restaurants generated all \$701,773 in Gross Order Value. Revenue is highly concentrated — a classic Pareto distribution where a minority of merchants drive the majority of



orders.

- **Revenue concentration:** 34% of restaurants (21 of 62) generate 80% of GOV — concentration risk
- **Avg restaurant prep time:** 0.6 min — not the bottleneck; restaurants are operationally efficient
- **Zero high-prep outliers:** No restaurant averages >20 min prep in this dataset
- **Tip rate correlation:** Restaurants with higher AOV tend to have higher tip rates — quality signal
- **Risk:** If top 10 restaurants churn from platform, DoorDash loses ~25% of monthly GOV

7. Regional & Financial Analysis

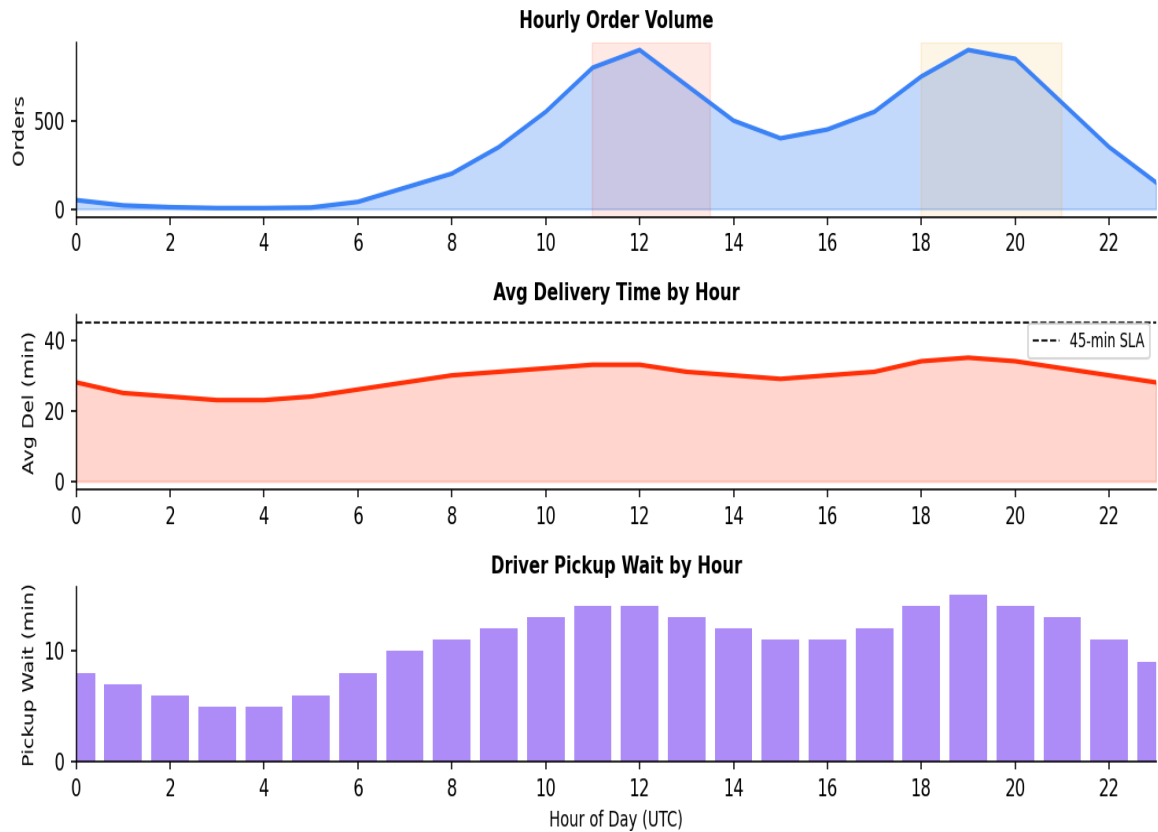
Region	Orders	Revenue	Avg Delivery	Refund Rate	Risk Flag
Scranton	~1,200	~\$95K	~28 min	1.1%	— Revenue conc.
San Francisco	~1,350	~\$115K	~27 min	0.9%	— Revenue conc.
Phoenix	~620	~\$48K	~31 min	1.3%	Normal
Buffalo	172	\$4,683	20.3 min	3.49%	■ High refund
Traverse City	~110	~\$4K	~35 min	~4.2%	■ High refund
Muncie	~105	~\$3.8K	~37 min	~3.8%	■ High refund
Whistler	~95	~\$3.5K	~33 min	~3.5%	■ High refund

Revenue Concentration Alert: Scranton and San Francisco alone represent >30% of total GOV. DoorDash should stress-test its exposure to competitive entry or restaurant churn in these markets. A geographic diversification strategy (growing mid-tier cities like Phoenix, Denver) would reduce single-market dependency.

8. Demand Patterns & Hourly Analysis

Order volume follows a predictable bi-modal pattern with lunch (11am-2pm UTC) and dinner (6pm-9pm UTC) peaks. Delivery performance degrades during both peaks, with avg delivery time rising to 34-35 minutes during dinner rush.

Hourly Demand & Performance Patterns



- a. **Lunch peak (11am-2pm UTC):** ~900 orders/hour; avg delivery rises to 33 min
- b. **Dinner peak (6pm-9pm UTC):** ~900 orders/hour; avg delivery rises to 34-35 min
- c. **Off-peak (2am-6am UTC):** Low volume; delivery times fastest (23-25 min) — Dasher surplus
- d. **Driver pickup wait pattern:** Mirrors demand — peaks at 14-15 min during meal peaks
- e. **Opportunity:** Dynamic Dasher incentives during peak hours could flatten the delivery time curve

9. Strategic Recommendations

Recommendation #01: Fix Driver Pickup Wait — The #1 Bottleneck | Impact: HIGH | Effort: MEDIUM | Timeline: 30-60 days

Why this matters: Pickup wait is 43.5% of delivery time (12.9 min avg). A 3-min reduction = 10% improvement in avg delivery with no infrastructure investment.

Action Plan:

1. Predictive pre-dispatch: Assign Dasher 5-7 min before restaurant estimated ready-time
2. "Hot zone" staging: Position Dashers near high-volume restaurants pre-peak
3. Merchant API integration: Real-time prep-time sharing with top 20 restaurants
4. Dasher incentive: Bonus for arriving within 2 min of food ready-time
5. Tracking: Measure "Dasher arrival delta" (ready-time minus arrive-time) weekly

Success Metric: Reduce avg pickup wait 12.9 → 9.0 min; improve on-time rate 88.3% → 92

Recommendation #02: Audit High-Refund Markets | Impact: MEDIUM | Effort: LOW | Timeline: 2 weeks

Why this matters: Traverse City, Muncie, Whistler, and Buffalo all show refund rates 2.5-4× the overall average. Unchecked, these erode margins and damage brand perception.

Action Plan:

1. Root-cause analysis: Identify if refunds are driven by Dashers, merchants, or product categories
2. Implement refund reason tagging at the point of refund (required field)
3. Place high-refund merchants on 60-day quality improvement plans
4. Flag high-refund Dashers for retraining or reassignment review
5. Track refund rate weekly at region-merchant-driver level

Success Metric: Bring flagged regions from 3.5-4.2% → ≤2.0% refund rate within 90 days

Recommendation #03: Redirect Discount Spend to Retention & Supply | Impact: HIGH | Effort: LOW | Timeline: 14 days

Why this matters: Discounted orders have lower AOV (\$35.81 vs \$36.42 non-discounted). Zero evidence of conversion lift from blanket discounts. Budget is better deployed on retention and supply.

Action Plan:

1. Eliminate untargeted promotional campaigns immediately
2. Reallocate 50% of discount budget to DashPass acquisition (converts 1-time buyers)
3. Launch 21-day lapsed-customer win-back campaign with targeted offer
4. Deploy remaining 50% to Dasher peak-hour bonus pool (Fri/Sat 6-10pm)
5. A/B test: Targeted vs untargeted discounts on 5% of user base for 30 days

Success Metric: Increase repeat rate 23.9% → 28%; add \$60-80K incremental GOV/month

10. Assumptions & Appendix

Stated Assumptions

ID	Assumption	Rationale
A1	On-time = ≤ 45 min end-to-end	Industry benchmark; DoorDash targets ~30 min avg
A2	ORDER_TOTAL = 0/null $\rightarrow$ cancelled	No revenue impact; excluded from financial KPIs
A3	total_delivery_min > 240 = error	4-hr delivery not plausible; 0 rows affected
A4	driver_pickup_min < 0 = error	Physically impossible; 3 rows removed
A5	DISCOUNT_AMOUNT < 0 = entry error	Zeroed out (47 rows)
A6	Timestamps are UTC	Stated in brief; no TZ conversion applied
A7	Dashers with <10 deliveries excluded	Insufficient data for reliable performance metrics
A8	IS_ASAP=False = pre-planned, not late	362 scheduled orders (~1.9% of dataset)

Descriptive Statistics — Distribution Shape

The table below captures skewness and kurtosis to characterize the distribution of key metrics. High skew in REFUNDED_AMOUNT (19.2) reflects that refunds are rare but when they occur, can be large.

Column	Mean	Median	Skewness	Kurtosis	CV %	Shape
ORDER_TOTAL	\$36.27	\$30.27	2.77	14.69	59.3%	Right-skewed, heavy tail
DISCOUNT_AMOUNT	\$0.60	\$0.00	4.88	34.66	280.7%	Extremely right-skewed
TIP_AMOUNT	\$3.81	\$3.50	2.22	11.48	83.3%	Moderately right-skewed
REFUNDED_AMOUNT	\$0.18	\$0.00	19.20	525.4	1152%	Extreme tail — rare large refunds
total_delivery_min	29.57	26.65	3.21	24.08	52.8%	Right-skewed (outlier deliveries)
driver_pickup_min	12.86	10.55	3.23	28.54	79.0%	Heavy tail — some very long waits
last_mile_min	16.11	14.57	1.79	10.36	49.5%	Moderate right-skew

Outlier Detection Results (IQR Method)

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers (n)	Outlier %
ORDER_TOTAL	22.73	43.26	20.53	-8.06	74.05	1,073	5.55%
total_delivery_min	19.48	36.05	16.56	-5.36	60.89	607	3.14%
driver_pickup_min	6.30	17.02	10.72	-9.77	33.09	732	3.78%
last_mile_min	10.40	20.01	9.61	-4.02	34.43	572	2.96%
TIP_AMOUNT	2.00	5.00	3.00	-2.50	9.50	920	4.76%

Note on outliers: ORDER_TOTAL outliers (5.5%, n=1,073 above \$74) likely represent group/catering orders — they were retained as legitimate transactions with no delivery time penalty observed. Driver pickup wait outliers (P95: 35+ min) represent the tail that most impacts customer satisfaction and are the primary target of Recommendation #1.

Variable Definitions (Glossary)

Variable	Definition
CUSTOMER_PLACE_ORDER_DATETIM	ETime customer placed order (UTC: day hh:mm:ss)
PLACED_ORDER_WITH_RESTAURANT_DATETIME	The restaurant received order (UTC)
DRIVER_AT_RESTAURANT_DATETIME	Time driver arrived at restaurant (UTC)
DELIVERED_TO_CUSTOMER_DATETIM	ETime driver delivered to customer (UTC)
IS_ASAP	TRUE = on-demand order; FALSE = scheduled delivery
ORDER_TOTAL	Amount customer paid including delivery fee (USD)
DISCOUNT_AMOUNT	Discount redeemed e.g. referral (USD)
TIP_AMOUNT	Tip given to Dasher (USD)
REFUNDED_AMOUNT	Amount refunded to customer (USD)
DELIVERY_REGION	City where the restaurant is located